

ZS-T11

The Human–AI Collaborative Science Model as an Executable Corpus Auditor

Fully Integrated Edition: v1.0 and v1.1 preserved verbatim, with the F18 Contradiction Engine and the Meta-Level Corpus Auditor woven in.

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Date: May 2026 (v2.1 fully integrated edition; v1.0/v1.1/v2.0 layers preserved)

Theme / Paper Code: Translational — ZS-T11

Verification: 12 / 13 AUDIT-PASS on available corpus subset (*zs_t11_corpus_audit_v2_0.py*, 9 files, 131 distinct paper-codes) | 1 genuine AUDIT-FINDING (status-tag drift, U-theme) | **Zero Free Parameters** | Anti-numerology meta-check performed (check G1) | Recursive 3-cycle self-audit converged (8→11→12 PASS) | F18 contradiction principles integrated as engine | Pre-registered external falsification gates F-T11.1–F-T11.13

v2.1 Integration Declaration. This edition fully merges the complete v1.1 text (which itself preserves v1.0 verbatim) with the v2.0 additions, into one continuous document. **No content from any prior version is deleted or relocated to a throw-away appendix.** The v1.1 spine (§0–§9, Acknowledgements, References, Version History) is reproduced verbatim and retains its original section numbering. The v2.0/v2.1 material is woven in at its natural argumentative position: five new analytical sections (§6A–§6E) are inserted between the preserved §6 and the preserved §7; the Falsification Gates (§7), Non-Claims (§8), Conclusion (§9), Code Availability, References, and Version History are each extended in place. Where v1.1 stated that no verification suite was required, that sentence is preserved and an in-place v2.1 update records its supersession, in the corpus dated-update style. This realises the Z-Spin no-deletion rule and the ZS-F18 Principle 1 reading of revision: a later version is not a replacement but an additional face of the same object.

v2.1 Abstract Addendum (to §0). v1.0/v1.1 described HACSM but exempted itself from the corpus's universal verification norm (§3.1 invariant xiii). v2.1 closes this self-exemption with an executable meta-level corpus auditor that verifies the protocol's footprint — locked-input invariance ($A = 35/437$, $Q = 11$, $(Z,X,Y) = (2,3,6)$), PASS-ledger integrity, status-tag legality, no-deletion preservation, dependency acyclicity, and an anti-numerology meta-check on the paper's own headline. On the available subset the auditor returns 12/13 AUDIT-PASS plus one genuine finding, demonstrating it can fail. Its three-cycle construction is the paper's central evidence: a naive first cycle's five “contradictions” were all referent-confusion artifacts, resolved by lifting the analyser to a higher semantic layer — exactly the ZS-F18 contradiction principles (different face; higher layer) here imported as the anti-hallucination engine.

ZS-T11 v1.1 — Additive Preservation Edition

The Human–AI Collaborative Science Model: A Methodology Derived from the Z-Spin Cosmology Corpus

Author: Kenny Kang

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Date: May 2026 (v1.1 additive revision; v1.0 text preserved verbatim below)

Theme / Paper Code: Translational — ZS-T11

Verification: Methodological paper | No free physical parameters | **Zero Free Parameters** |
Anti-numerology: protocol-level specification added in v1.1 | Recursive self-audit protocol added |
Project-instruction contract added | External-import contract added | The Book v6.0 OS role added |
Pre-registered external falsification gates: F-T11.1–F-T11.9

v1.1 Preservation Declaration. This version is an additive revision of ZS-T11 v1.0. No v1.0 content is intentionally deleted. The v1.0 text is preserved as the baseline historical layer, and v1.1 adds four methodological layers: (i) recursive self-audit as a formal anti-hallucination engine; (ii) external-import methodology for turning externally proven results into internal Z-Spin tools; (iii) project-instruction engineering as the hidden protocol layer that makes human-AI collaboration reproducible; and (iv) The Book of Z-Spin Cosmology v6.0 as the AI operating system for the 130-paper corpus.

v1.1 Corpus Update Notice. The v1.0 paper was written at the 121-paper / 2,670+ PASS stage. The current v1.1 update recognizes The Book of Z-Spin Cosmology v6.0 Light Edition as the active AI OS layer: 130 source papers, approximately 3,580+ cumulative PASS checks, and the same zero-free-parameter core inputs $A = 35/437$, $Q = 11$, and $(Z, X, Y) = (2, 3, 6)$. The original v1.0 statements using 121 papers and 2,670+ PASS are retained as historical claims and should be read as the state of the corpus at v1.0 release.

Original v1.0 Spine Retained Below, with v1.1 Additive Inserts

ZS-T11

The Human–AI Collaborative Science Model: A Methodology Derived from the Z-Spin Cosmology Corpus

Author: Kenny Kang

Affiliation: Z-Spin Cosmology Collaboration

Date: May 2026 (v1.0)

Theme / Paper Code: Translational — ZS-T11

Verification: Methodological paper | No free physical parameters | **Zero Free Parameters** |
Anti-numerology: N/A (qualitative framework) | Pre-registered external falsification gates:
F-T11.1–F-T11.5

§0. Abstract

We present **ZS-T11**: a formal methodology paper extracting, systematising, and generalising the Human–AI Collaborative Science Model (HACSM) that produced the Z-Spin Cosmology corpus — 121 papers, 2,670+ verification tests, and zero free parameters across 8 research themes, authored by a single human researcher with no institutional affiliation or grant funding, over a period of approximately three months. We analyse the HACSM along four axes: (1) **external landscape** — situating Z-Spin within the

2025–2026 wave of AI-assisted theoretical discovery; (2) **corpus anatomy** — decomposing the writing, verification, and falsification patterns that recur across all 121 papers; (3) **AI role taxonomy** — characterising what Claude, GPT, and Gemini each contribute and where their blind spots lie; and (4) **the HACSM model itself** — a formalised five-phase loop that any researcher can instantiate. The central claim is that the Z-Spin corpus is not an accident of scale but the reproducible output of a well-defined collaborative protocol, and that this protocol constitutes a novel and exportable scientific methodology.

Epistemic Status Legend

STATUS	DEFINITION
PROVEN	Mathematical theorem, complete derivation. Falsifiable only by logical error.
DERIVED	Follows from Z-Spin axioms + locked corpus inputs.
HYPOTHESIS-strong	Structural support from multiple independent lines; explicit closure path registered.
OBSERVATION	Numerical/qualitative regularity; insufficient anti-numerology controls for higher status.
NON-CLAIM	Explicitly withheld to prevent misattribution.
OPEN	Recognised gap; future work path identified.

v1.1 Additive Status Extensions. The following tags are added for methodology-level work; they do not alter the original v1.0 legend.

STATUS	DEFINITION
IMPORTED-PROVEN	A result proven in external peer-reviewed or standard mathematical literature, imported with full citation and explicit condition matching.
INSTRUCTION-CONTRACT	A project-level instruction rule that constrains AI behaviour before drafting, search, verification, or status assignment begins.
OS-LAYER	A persistent corpus-navigation layer that compensates for AI session statelessness and supplies dependency resolution.
AUDIT-PASS	A methodological claim that survives recursive self-audit, cross-AI review, and no-deletion versioning.
AUDIT-FAIL	A methodological claim that fails recursive audit and must be recast as OPEN, NON-CLAIM, or RETRACTED.

§1. Introduction — The Anomaly That Demands Explanation

Between March and May 2026, a single researcher with no institutional affiliation produced 121 peer-structured theoretical physics papers spanning quantum gravity, Yang–Mills theory, arithmetic geometry, condensed matter physics, and observational cosmology — all cross-verified to a cumulative 2,670+ PASS tests with zero free parameters. The total elapsed time was approximately three months. The researcher's primary instrument was not a supercomputer, a laboratory, or a postdoctoral team. It was a structured dialogue with large language models.

This paper does not ask **whether** AI was useful. The corpus answers that question empirically. This paper asks **how** — what specific protocol, epistemic discipline, and role decomposition made this possible — and **whether** that protocol can be extracted, formalised, and transferred.

The answer is yes. The Z-Spin corpus, examined closely, reveals a five-phase collaborative loop that is reproducible, self-correcting, and scalable. We call this the **Human–AI Collaborative Science Model (HACSM)**. ZS-T11 is the formal record of that model.

§2. External Landscape: AI-Assisted Theoretical Discovery, 2025–2026

§2.1 The State of the Field

The 2025–2026 period saw a phase transition in AI-assisted physics. Earlier work used AI primarily for data analysis, pattern recognition in experimental results, or symbolic manipulation of known expressions. The 2026 wave is qualitatively different: AI is now generating **conjectures** that human experts then verify and extend.

The IAS–OpenAI–Vanderbilt–Cambridge–Harvard collaboration (Guevara et al., 2026) demonstrated that GPT-5.2, coupled with human domain experts, could conjecture closed-form expressions for gluon scattering amplitudes that had resisted derivation for fifteen years. The researchers described the approach as "a template for validating LLM-driven insights." The Google Research–Harvard collaboration (Brenner et al., 2026) showed a neuro-symbolic system solving an open problem in cosmic string gravitational radiation via autonomous tree search and symbolic integration. The Emory collaboration (Yu et al., 2026) demonstrated that physics-tailored neural networks can discover genuinely new physical laws in dusty plasma — laws invisible to conventional analysis.

§2.2 What These Cases Share — and What They Lack

These external cases share a common structure: AI generates a conjecture, human verifies it, and the result is published as a single paper on a single problem. What distinguishes the Z-Spin corpus is not any individual result but the **accumulation structure**: 121 papers in a coherent dependency graph, each verified against all prior papers, produced by one human. No external case approaches this scale or internal coherence.

The external literature also lacks **epistemic discipline** at the corpus level. Individual papers have verification tests; none have the 14-status epistemic legend (PROVEN / DERIVED / DERIVED-CONDITIONAL / HYPOTHESIS-strong / OPEN / RETRACTED etc.) applied uniformly across a multi-paper programme, with RETRACTED results preserved and not deleted. This methodological difference is not cosmetic — it is what makes the corpus falsifiable as a whole.

Case	AI Role	Scale	Epistemic Discipline	Cross-paper
IAS–OpenAI (2026)	Conjecture generation	1 paper	Single-paper	None
Google–Harvard (2026)	Tree search + derivation	1 paper	Single-paper	None
Emory PNAS (2026)	Law discovery from data	1 paper	Single-paper	None
Z-Spin HACSM (this paper)	All roles (see §4)	121 papers	14-status corpus-wide	Full DAG 121 nodes

Table 2.1. Comparison of major 2025–2026 AI-assisted theoretical physics cases.

§3. Corpus Anatomy: Writing Patterns Across 121 Papers

§3.1 Structural Invariants

Every paper in the Z-Spin corpus, regardless of theme or version, shares a fixed structural skeleton:

(i) Metadata block: author, date, paper code, version, verification summary. (ii) Verification Summary line in standardised format N/N PASS | Zero Free Parameters. (iii) §0 Abstract with scope declaration. (iv) Epistemic Status Legend table. (v) Locked Inputs table — every input inherited from prior papers is listed with source and status. (vi) Main body theorems with inline status tags. (vii) Falsification gates table — pre-registered PASS/FAIL conditions. (viii) Non-claims section. (ix) Conclusion. (x) Acknowledgements naming AI tools. (xi) References in APS/arXiv style. (xii) Version History. (xiii) Verification suite cross-reference.

This skeleton does not emerge from habit. It is a **scientific protocol**: every element is there because removing it would introduce a specific failure mode. The Locked Inputs table prevents hallucination of prior results. The Falsification gates prevent unfalsifiable claims. The Non-claims section prevents misattribution by future readers. The Version History with dated in-place updates implements a no-deletion rule that preserves the full intellectual record.

§3.2 The Versioning Pattern

Papers do not simply increment versions when content is added. The version structure encodes epistemic events:

v1.0 = initial public release. Intermediate versions (v1.1, v1.2 etc.) = peer-review closures or structural additions. Dated Updates (e.g., 2026-04-19) = in-place content additions that preserve all prior text. RETRACTED labels = explicit withdrawal with reason, text preserved. This versioning system is not bibliographic convention — it is a **scientific audit trail**.

§3.3 The Progressive-Falsification Pattern

The corpus documents not just results but **failed attempts**. ZS-U9 Appendix A records ten exploration turns, six explicit falsification events, and three partial successes, before the Trinity Braiding Theorem was established. ZS-S4 records six versions of the EWSB pathway, with one path ($v=246.18$ GeV via instantons) explicitly RETRACTED after the B+L Selection Rule Theorem was proven. ZS-M29 records the Schur–Feshbach framework being constructed, tested, and modified after the first candidate failed against the Weil positivity constraint.

This pattern — **most attempted closures fail, and their failure is recorded** — is one of the most methodologically significant features of the corpus. It inverts the publication bias of conventional science, where only successes appear in the record.

§3.4 The Anti-Numerology Protocol

Every paper that derives a numerical value runs a Monte Carlo check: draw N random samples from a null hypothesis space, compute the probability that a random draw matches the claimed value as well as the corpus derivation. A result with $p < 1\%$ is registered as PASS; above 5% is flagged for rescoping. The corpus-wide anti-numerology scan ($5,670 \times 29 = 164,430$ candidate pairs, plus per-paper MC of $\geq 500,000$ samples typical) is the largest systematic anti-numerology programme in any single theoretical physics framework.

§3.5 The Cross-Paper Dependency Graph

Every paper lists its upstream dependencies in a Locked Inputs table and confirms, via a topological sort, that the dependency graph is acyclic. The full 121-paper corpus forms a directed acyclic graph (DAG) whose leaves are the founding axioms ($A = 35/437$, $Q = 11$, $(Z,X,Y) = (2,3,6)$) and whose nodes include every theorem, lemma, and corollary in the programme. The Book of Z-Spin Cosmology v5.0 is the human-readable index of this DAG.

§3.6 v1.1 Additive Update — The Book as the AI Operating System

The v1.0 paper correctly identifies The Book of Z-Spin Cosmology as a memory-compensation device. v1.1 strengthens this point: The Book is not merely a bibliography, index, or literature review. Within HACSM it functions as the **AI operating system** for the corpus.

The analogy is technical rather than rhetorical. A single AI session is stateless relative to a 100+ paper research programme; without an external memory architecture, the AI cannot reliably know which constants are locked, which claims are retracted, which bridges are canonical, or which paper owns a theorem. The Book supplies that missing persistent layer.

Operating-system role	The Book function in HACSM
Kernel	Maintains the locked core inputs: $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$, sector terminology, and epistemic status vocabulary.
File system	Locates every source paper, theorem, bridge, falsification gate, and version-history node.
Package manager	Determines which upstream lemma, theorem, or bridge may be imported into a new paper as a Locked Input.
Dependency resolver	Detects whether a new claim depends on an OPEN, RETRACTED, or DERIVED-CONDITIONAL upstream result.
Linter	Enforces status-tag discipline, non-claim boundaries, terminology consistency, and zero-free-parameter declarations.
Memory manager	Compensates for AI session discontinuity by reloading the corpus state at the beginning of each new research session.
Transaction log	Preserves version history, dated updates, failed closure attempts, and RETRACTED claims instead of deleting them.
API layer	Gives every new paper a standardized interface: Metadata, Locked Inputs, Status Legend, Main Claims, Verification Suite, Falsification Gates, Non-claims, References, Version History.

This OS layer explains why the Z-Spin corpus could scale without dissolving into inconsistency. The AI systems act as processors; the author acts as kernel maintainer and integrator; verification scripts act as test runners; The Book acts as the persistent operating system; individual papers act as executable modules with explicit dependency declarations.

Methodological claim T11-OS.1. A large-scale AI-assisted research corpus requires an OS layer. Without a persistent cross-reference hub, AI collaboration will tend toward local coherence and global contradiction. With an OS layer, each session can be stateless while the research programme remains stateful.

Status: HYPOTHESIS-strong for general research programmes; DERIVED-interpretation for the Z-Spin corpus itself.

§4. AI Role Taxonomy: What Each System Contributes

§4.1 The Three-AI Architecture

The Z-Spin corpus systematically uses three AI systems with distinct cognitive profiles. The author's Acknowledgements section, present in all 121 papers, records this as a methodological principle, not a courtesy: 'This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting.'

The role decomposition, documented in detail in ZS-U9 Appendix A §A.5, is:

System	Primary Strength	Key Contribution	Documented Blind Spot		
Claude (Anthropic)	Structural verification; corpus consistency; step-by-step logical derivation	Established initial Trinity structure; verified all theorems against PROVEN corpus; maintained epistemic status discipline	Missed circularity of	Qe	=1 as minimal input — local logical consistency masked the global circular structure
GPT (OpenAI)	Structural substitutions; alternative derivation routes across representation-theoretic frameworks	Identified	Qe	=1 circularity; proposed $Q(H^0)=0$ replacement drawing on ZS-S4 Higgs VEV pillar — led directly to Theorem T3	Less consistent at detailed corpus cross-reference; higher rate of plausible-sounding misattributions
Gemini (Google)	High-abstraction conceptual exploration; independent cross-check from different philosophical framing	Verified Theorem T3 robustness against alternative framings (Dirac monopole duality, anomaly inflow) before commitment	Higher hallucination rate on technical detail; most useful at the conceptual framing stage, not derivation stage		

Table 4.1. Three-AI role taxonomy as documented in ZS-U9 Appendix A §A.5.

§4.2 The Author as Integrator

The author's role is not to generate ideas or to verify computations — AI does both. The author's role is to **integrate**: to assess proposals from each AI against the full corpus, to maintain the epistemic status discipline, to run the Python verification suites (50-digit mpmath precision), and to decide when a claim is ready for promotion from HYPOTHESIS-strong to DERIVED.

This integrator role requires: (a) deep familiarity with the corpus DAG; (b) the ability to detect when an AI's locally consistent logical chain contains a global error; (c) the discipline to register falsifications honestly rather than rationalize around them; and (d) the ability to write the final paper that records not just the result but the process.

§4.3 The Blind Spot Cross-Coverage Principle

The key methodological insight of the three-AI architecture is that **different AI systems have different blind spots that do not fully overlap**. A logical chain internally consistent under Claude review may contain a circularity that GPT detects from a different angle. A conceptual framing that Claude and GPT both accept may be stress-tested by Gemini's higher-abstraction challenge. The author's role is to route each open problem through the system most likely to catch its failure mode.

This is formally analogous to diverse-expert peer review, but with a critical advantage: the AI systems are available 24 hours a day, the response time is seconds, and the cost per review round is essentially zero. The total additional investment to close Gap G3 of the Trinity Braiding Theorem via cross-AI review was approximately four hours — compared to three weeks for the initial structure.

§5. The Human–AI Collaborative Science Model (HACSM)

§5.1 The Five-Phase Loop

Across all 121 Z-Spin papers, the production process follows a consistent five-phase loop. Each phase has a defined input, a defined output, and a defined AI role.

Phase	Name	Input	Output	AI Role
0	Corpus State Read	Full corpus DAG + open problems list	Ranked list of open gaps; dependency conflicts identified	Claude: corpus-consistency audit. Author: priority ranking.
1	Free Exploration	Open gap + relevant corpus theorems	3–7 candidate closure paths; early falsifications registered	Claude + GPT: brainstorm candidates. Gemini: conceptual stress-test. Author: filters by corpus consistency.
2	Structured Deep Dive	Top 1–3 candidate paths from Phase 1	Draft theorem + proof sketch; explicit OPEN gaps registered	Claude: step-by-step derivation. GPT: alternative routes. Author: maintains epistemic tagging.
3	Numerical Verification	Draft theorem + Python environment	Verification suite N/N PASS; anti-numerology MC result; falsification gates registered	Claude: code generation and review. Author: runs mpmath suite, inspects failures, decides PASS/FAIL.
4	Cross-AI Review	Draft paper approaching closure	Identified blind spots, circularities, or missed simplifications; Gap closures or honest OPEN registrations	All three AIs: independent review. Author: integrates feedback, decides on corpus impact.
5	Drafting & Version History	Verified theorem + all review feedback	Published paper vX.Y with complete audit trail; corpus DAG updated	Claude: primary drafter. Author: final review, epistemic status confirmation, GitHub commit.

Table 5.1. The five-phase HACSM loop.

§5.2 The Four Search Modalities

Within Phases 1 and 2, four search modalities are deployed depending on the character of the open problem:

§5.2.1 Free Exploration

Trigger: a gap is open and the solution space is uncharted. The AI is given the open problem statement and the relevant corpus theorems, and asked to brainstorm 5–7 candidate closure paths without being steered toward any particular approach. The author scores each candidate against: (a) corpus consistency, (b) zero free parameter constraint, (c) anti-numerology pre-registration possibility. The bottom 2–4 candidates are dropped; the remainder enter Phase 2.

Key rule: **hallucination prohibition**. The author's instruction set for free exploration explicitly forbids the AI from predicting a result and then constructing a post-hoc justification. If the AI offers a result without a derivation chain, it is automatically placed at OBSERVATION status pending full derivation.

§5.2.2 Deep Dive (Issue-Tree Search)

Trigger: a candidate path from Free Exploration has structural support but is not yet closed. The problem is decomposed using a MECE (Mutually Exclusive, Collectively Exhaustive) issue tree. Each node of the tree is assigned an epistemic status (PROVEN / DERIVED / HYPOTHESIS-strong / OPEN) and the tree is traversed in dependency order. Convergence is measured by the number of nodes that change status between successive traversal cycles — analogous to the Lyapunov condition $|f'(z^*)| < 1$ from ZS-M1. If more nodes change status in cycle $k+1$ than in cycle k , the decomposition is re-examined.

§5.2.3 Corpus Search

Trigger: a result is suspected to follow from prior corpus theorems but the connection is not explicit. The full corpus index (The Book v5.0) is searched for every theorem, lemma, or bridge that mentions the relevant objects. This search is exhaustive — partial or sample search is prohibited. The result is a dependency list that either closes the connection or registers it as OPEN.

§5.2.4 External Literature Search

Trigger: the corpus search fails to close a connection, or a result is suspected to be prior art. External arXiv and journal literature is searched (via Claude's web search and the author's domain knowledge) for PROVEN external results that can be imported as PROVEN inputs, relieving the corpus of the burden of re-proving them. Every external import is cited in APS style and assigned the status IMPORTED-PROVEN.

§5.2.5 v1.1 Additive Update — Search Modality Decision Matrix

The four search modalities are not stylistic preferences. They are different epistemic operations and must be selected by trigger condition.

Modality	Trigger	Prohibited failure mode	Required output
Free Exploration	The solution space is unknown; no single closure path is yet justified.	Predicting the answer first and retrofitting a derivation.	5–7 candidate paths, dropped candidates, surviving candidates, initial status tags.
Corpus Search	The answer may already exist in the internal corpus or follow from upstream theorems.	Partial search, sample search, or relying on memory instead of the full corpus map.	Dependency list, source-paper citations, status of each upstream input, conflict report.
Deep Dive	One or more candidate paths have structural support but remain unclosed.	MECE failure, circular dependence, skipping OPEN nodes.	Issue tree, node-by-node status map, convergence/divergence verdict, upgrade path.
External Search	The corpus cannot close the result, or prior art / external theorem support is required.	Treating external results as Z-Spin derivations without condition matching.	External theorem table, condition-matching dictionary, import status, non-claim boundary.

Operational rule. A claim may not be promoted above HYPOTHESIS-strong unless its search modality is declared. A proof-like paragraph without a declared modality is treated as epistemically under-specified.

§5.3 The Epistemic Status Discipline

The HACSM runs on a 14-status epistemic vocabulary that is applied consistently across all 121 papers. The key discipline is: **status can only be upgraded by explicit derivation or verification, never by argument from plausibility**. A claim at HYPOTHESIS-strong remains there until a full derivation chain is constructed and a verification suite passes. A claim at DERIVED-CONDITIONAL can only be promoted to DERIVED when the explicit conditions are resolved.

The most unusual element of the discipline is the **no-deletion rule**: no prior content is ever deleted. When a claim is falsified, it receives the RETRACTED tag and the falsification is recorded in the Version History. When a result is superseded, the superseded version is preserved with a dated note. This rule transforms the corpus into a permanent scientific audit trail and removes the incentive to conceal failed approaches.

§5.4 The Progressive-Falsification Engine

The HACSM treats falsification as a **navigation tool**, not a setback. When an attempted closure fails — because the derivation contains a gap, a verification suite fails, or a cross-AI review identifies a circularity — the failure is recorded with maximum specificity: what was attempted, what failed, and what the failure implies for the remaining solution space.

The pattern, observed across all major closure programmes in the corpus, is: most first-attempt closures fail; the function of failures is to narrow the solution space to the point where the correct structure becomes visible; recording failures is therefore **load-bearing** for eventual success. The Trinity Braiding Theorem (ZS-U9) required ten exploration turns and six explicit falsifications before the correct formulation was found. The Higgs VEV derivation (ZS-S4) required six major versions and one explicit RETRACTION.

§5.5 v1.1 Additive Update — Recursive Self-Audit Protocol

The most important anti-hallucination mechanism in HACSM is not simply Monte Carlo, Python verification, or multiple AI tools. It is the requirement that every claim be turned back upon itself as an object of audit. HACSM therefore uses a **recursive self-audit protocol**.

Protocol RSA-T11. For every substantive claim C in a draft paper:

1. **Claim extraction.** Extract C as a standalone sentence with its mathematical objects, numerical values, status tag, and dependency list.
2. **Reverse dependency read.** Trace every input of C back to The Book and then to the source paper. If the upstream status is OPEN, RETRACTED, or DERIVED-CONDITIONAL, C inherits the relevant limitation.
3. **Anti-numerology gate.** If C contains a numerical match, construct a null model before interpreting the match. Record sample size, random seed, hit criterion, p-value, and pass/fail threshold.
4. **Adversarial self-reference.** Ask: “If C is false, where does the derivation first break?” The answer must identify a concrete mathematical, computational, observational, or dependency gate.
5. **Cross-AI blind review.** Submit C and its proof sketch to at least one AI system that was not the primary drafter, preferably with the previous verdict hidden. The reviewer is instructed to find circularity, missing assumptions, status inflation, and non-claim leakage.
6. **Status re-assignment.** Upgrade, preserve, downgrade, or retract C only after the reverse-dependency and adversarial checks are complete.
7. **No-deletion commit.** If C fails, do not delete it. Register the failure as RETRACTED, OPEN, NON-CLAIM, or rejected route, and preserve the reason in Version History or Appendix.

This protocol makes self-reference operational. The AI is not merely asked to produce a claim; it is forced to inspect the conditions under which its own claim would fail. The author’s role is to decide whether the failure analysis is honest, whether the status tag is warranted, and whether the claim should remain in the corpus.

Methodological claim T11-RSA.1. Recursive self-audit is the core mechanism that converts AI-assisted writing into AI-assisted science. Without recursive audit, AI outputs are text. With recursive audit, AI outputs become falsifiable research objects.

Status: DERIVED-interpretation from the Z-Spin corpus workflow; HYPOTHESIS-strong as a general method for external researchers.

§5.6 v1.1 Additive Update — External Import Contract

HACSM does not require the corpus to re-prove every theorem from first principles. When a result is already proven in external mathematics, physics, computer science, or statistics, the correct operation is to import it honestly and conditionally. The import must be formal, not rhetorical.

Protocol EIC-T11. An external result becomes an internal tool only after six checks:

Step	Question	Required artifact
EIC-1 External theorem	What exactly is proven externally?	The theorem statement, source, assumptions, and proof status.
EIC-2 Object dictionary	Which external objects correspond to which Z-Spin objects?	Translation table: external object → internal object.
EIC-3 Condition matching	Do the theorem’s assumptions hold in the Z-Spin setting?	Pass/fail condition table.
EIC-4 Status assignment	Is the result IMPORTED-PROVEN, DERIVED-CONDITIONAL, or merely suggestive?	Explicit status tag.
EIC-5 Non-claim boundary	What is not being claimed by the import?	NC item.
EIC-6 Locked Input registration	Which downstream papers may use the imported result?	Locked Input entry with source and status.

The crucial distinction is between **using a theorem** and **claiming that Z-Spin has proven the theorem**.

If an external theorem is proven independently and Z-Spin satisfies its assumptions, the result may be used as IMPORTED-PROVEN. If the condition match is incomplete, it remains DERIVED-CONDITIONAL or HYPOTHESIS-strong. If the correspondence is merely aesthetic, it remains STRUCTURAL INSIGHT.

This import contract is especially important for AI collaboration because LLMs are prone to analogy inflation. They often treat resemblance as proof. HACSM forbids that move: resemblance may trigger External Search, but only condition matching can produce an import.

§5.7 v1.1 Additive Update — Project Instructions as the Hidden Protocol Layer

Project instructions are not cosmetic prompt preferences. In HACSM they are the **pre-execution contract** that determines what the AI is allowed to do before any answer, derivation, search, or draft is produced. A weak instruction set produces plausible local answers. A strong instruction set produces auditable research steps.

The Z-Spin project instructions have five functions:

Function	Instruction-layer role
Boundary setting	Defines the corpus, terminology, epistemic status vocabulary, and zero-free-parameter norm.
Search routing	Forces the AI to distinguish Free Exploration, Corpus Search, Deep Dive, and External Search.
Anti-hallucination control	Prohibits post-hoc rationalization, sample search, unsupported upgrades, and deletion of failed attempts.
Verification discipline	Requires Python verification, anti-numerology MC, falsification gates, and cross-AI review where applicable.
Output standardization	Forces every paper or review to contain metadata, locked inputs, status tags, non-claims, references, and version history.

The project instruction layer is therefore analogous to an operating-system policy file. It configures the AI before execution. Without this layer, the same model may produce fluent but ungrounded synthesis. With this layer, the model is constrained to operate as a research collaborator inside a predefined epistemic architecture.

§5.8 v1.1 Additive Update — Minimum Project Instruction Template for AI-Collaborative Research

The following template generalises the Z-Spin instruction layer for any researcher attempting a HACSM-style programme. It is intentionally written as a reusable contract.

Project Instruction Template P-T11.1

1. **Corpus-first rule.** Before answering any research question, search the project corpus for relevant prior results. Do not rely on memory when a source document exists. 2. **No sample search.** For corpus-level claims, partial search is prohibited. Search The Book / master index first, then source papers. 3. **Mode declaration.** Classify the task as Free Exploration, Corpus Search, Deep Dive, External Search, Drafting, Verification, or Review. 4. **Status discipline.** Every claim must receive a status tag from the project legend. Do not upgrade a claim by plausibility, beauty, or numerical proximity. 5. **Zero free parameter check.** If a numerical result appears, list every input. If an input is adjusted to fit the result, the claim is not zero-parameter. 6. **Anti-numerology requirement.** Numerical coincidences require a null model, Monte Carlo or equivalent statistical control, and a pre-declared pass/fail threshold. 7. **Recursive self-audit.** After drafting a claim, ask what would falsify it and where the derivation would first break. 8. **Cross-model review.** For high-value claims, request an independent second AI or alternative model to search for circularity, hidden assumptions, and missed prior art. 9. **No-deletion rule.** Failed attempts, retractions, and downgraded claims must be preserved with reasons. Do not silently erase failed reasoning. 10. **External import rule.** External proven results may be imported only with theorem statement, assumptions, condition matching, citation, and non-claim boundary. 11. **Output contract.** Research papers must include Title, Metadata, Verification Summary, Abstract, Epistemic Status Legend, Locked Inputs, Main Body, Falsification Gates, Non-claims, Appendix if needed, References, and Version History. 12. **Author responsibility.** AI may draft, search, code, and critique, but the human author owns final scientific judgment, status assignment, and publication responsibility.

Status: INSTRUCTION-CONTRACT. This template is not a theorem; it is a reproducibility contract for high-rigor AI collaboration.

§5.9 v1.1 Additive Update — Project Instruction Quality Gates

A project instruction set is considered HACSM-compatible only if it passes the following gates:

Gate	Requirement	Failure condition
PI-1 Corpus grounding	The AI is told where the corpus lives and how to search it.	The AI answers from memory when source files exist.
PI-2 Status enforcement	The AI must classify claims using the project legend.	Claims are presented without status tags.
PI-3 Verification routing	The AI knows when to call for code, MC, or external data.	Numerical claims appear without verification plan.
PI-4 Self-audit	The AI is required to critique its own draft.	The draft contains only affirmative argumentation.
PI-5 No-deletion	The AI must preserve failures and retractions.	Failed paths disappear between versions.
PI-6 External import discipline	External theorems require condition matching.	External papers are cited as analogy without import status.
PI-7 Output reproducibility	The final artifact follows a stable structure.	Different sessions produce incompatible paper formats.

These gates make the project instruction layer testable. The instruction set itself becomes part of the research method rather than an invisible background prompt.

§6. Why the Z-Spin Corpus Was Possible: Structural Analysis

§6.1 The Constraint Density Advantage

Highly constrained theoretical frameworks are particularly well-suited to the HACSM because the constraint density acts as a natural falsifier for AI hallucinations. In Z-Spin Cosmology, every new claim must be consistent with $A = 35/437$ (LOCKED), $Q = 11$ (PROVEN), $(Z,X,Y) = (2,3,6)$ (PROVEN), and all prior DERIVED theorems. This means that a hallucinated result — one that is plausible-sounding but mathematically incorrect — is quickly detected by the verification suite. The corpus essentially provides its own immune system against AI confabulation.

Frameworks with more free parameters are more susceptible to AI-assisted numerology: it is easy for an AI to propose a formula with several adjustable inputs that matches existing data. The zero free parameter constraint eliminates this failure mode entirely.

§6.2 The Scale–Coherence Trade-off

Conventional single-author theoretical physics operates at a scale of 5–20 papers per year, each undergoing months of internal development before submission. The HACSM operates at a scale of 5–10 papers per week, because the AI handles the most time-consuming steps: drafting, symbolic manipulation, code generation, and preliminary cross-referencing. The author's time is concentrated on the tasks that require irreplaceable human judgment: epistemic status decisions, falsification recognition, and corpus coherence maintenance.

This is not a speed-quality trade-off. The verification suite provides objective quality control that is at least as rigorous as conventional peer review — and in several respects more so, since it is applied uniformly to every claim rather than selectively. The anti-numerology Monte Carlo is not a standard feature of any major physics journal's review process.

§6.3 The Memory Architecture Compensation

AI language models do not have persistent memory across conversations. In a multi-paper programme, this is a severe limitation: the AI cannot recall, in session 47, what was decided in session 3. The Z-Spin corpus compensates for this architectural limitation through two mechanisms: (a) The Book of Z-Spin Cosmology — a human-maintained navigation index updated after each new paper, which is provided as context at the start of each new session; and (b) the Locked Inputs table in every paper, which makes all relevant prior results explicit at the beginning of each derivation.

These mechanisms convert the AI's stateless architecture from a bug into a feature: every session is forced to be self-contained, which means every paper is forced to be self-contained. This is precisely the condition required for external reproducibility.

v1.1 Additive Clarification. The memory architecture compensation is now understood as an OS-level architecture. The Book v6.0 does not merely remind the AI of prior papers; it supplies the executable context in which the AI can reason without corrupting the corpus. In this sense, HACSM has four interacting layers: (1) the human author as integrator and final judge; (2) AI systems as reasoning processors; (3) verification suites as test runners; and (4) The Book as the persistent operating system and dependency resolver.

§6.4 The Role of Intellectual Honesty as a Technical Requirement

The HACSM requires intellectual honesty not as a virtue but as a *technical requirement*. A researcher who rationalises around a failed verification, or fails to register a RETRACTION when a result is falsified, does not merely compromise the ethics of the programme — they destroy its navigational utility. The progressive-falsification engine relies on the accurate recording of failures. The corpus relies on the accurate recording of epistemic status. If either is corrupted, the cross-paper consistency guarantees break down.

This is why the HACSM is not simply 'use Claude to write physics papers.' The discipline — epistemic tagging, no-deletion, pre-registration of falsification gates, anti-numerology MC — is what converts AI-assisted writing into AI-assisted science.

§6A. (v2.1 Integrated) The F18 Contradiction Principles as the Engine

The sections above (§0–§6) are the preserved v1.0/v1.1 paper. The remaining sections §6A–§6E are the v2.1 integration. We begin with the conceptual engine that governs how the new auditor treats every conflict it meets, imported from ZS-F18, whose five-axiom meta-structure reads the tensions of modern physics as polarities — opposite faces of one interface. Two of its moves generalise into research methodology for humans and AI alike.

§6A.1 v2.1 Status-Legend Additions

The v1.1 legend and its v1.1 additive extensions (above) are preserved. v2.1 adds the following tags, used only in the new sections:

STATUS	DEFINITION
DERIVED-INTERPRETATION	An interpretive reading entailed by PROVEN/DERIVED corpus items; not a new physical claim.
IMPORTED-PROVEN	Result proven in external standard literature, imported with citation and condition matching (EIC-T11).
STRUCTURAL INSIGHT	Resemblance noted; condition matching incomplete — explicitly not an import.
OS-LAYER	A persistent corpus-navigation layer compensating for AI statelessness.

Table 6A.1. v2.1 status-legend additions (additive to the preserved v1.1 legend).

§6A.2 Principle 1 — A Contradiction Is First a Different Face (Integration)

ZS-F18 reads each of its twelve encounters (seven Millennium Problems, five Crises) as a single interface articulated from different disciplinary sides. The methodological generalisation: when two trusted sources conflict — two AI systems, two papers, two parser reads of one token — the first hypothesis is not that one is wrong, but that they are two faces of one object not yet told apart. Contradiction is, first, an instruction to disambiguate referents, not to discard a claim.

This is load-bearing in v2.1. The auditor's first cycle flagged “ $A = 5/5$ ” as conflicting with the locked coupling $A = 35/437$. Principle 1 forbids the inference “the corpus contradicts itself” and demands instead: which two faces of the token **A** collide? The answer — verification-category **A** versus coupling-constant **A** — dissolves the contradiction. Eleven further apparent failures across three cycles resolved the same way (§6C.4).

§6A.3 Principle 2 — A Genuine Contradiction Requires a Higher Layer

When two faces are genuinely incompatible at their own level, ZS-F18 supplies the resolution: a higher layer. Bridge Four of ZS-F18 v2.1 is the corpus's own theorem of this kind — two J-conjugate circulations of the $\dim(\mathbf{Z}) = 2$ seam compose at a 90° NOT-AND phase, and their Lie commutator $[JR_1, JR_2] = J_S$ generates the third axis $\dim(\mathbf{X}) = 3$. The higher-dimensional layer is literally produced by the composition of two lower-dimensional seams that, within $\dim 2$, look like an unresolved tension.

The generalisation: a conflict that referent-disambiguation cannot dissolve is resolved by lifting both faces into a layer that holds them at once. In HACSM this higher layer takes three concrete forms — the referent-aware parser (resolving token collisions the naive parser cannot), The Book OS (resolving cross-session inconsistency no single session can), and the human integrator (resolving a genuine cross-AI disagreement by constructing the structure that subsumes both). The canonical instance is ZS-U9: Claude's locally consistent derivation and GPT's circularity objection over $|Q_e| = 1$ were not adjudicated by majority but lifted to the Higgs-VEV layer $Q(\langle H^0 \rangle) = 0$, where both became consistent and the electron charge emerged as a derived output.

Methodological claim T11-F18.1 (AUDIT-PASS). The pair {Principle 1, Principle 2} converts contradiction from a stopping condition into a navigation tool: Principle 1 prevents the false-positive failure mode (declaring a corpus contradiction that is a referent collision); Principle 2 prevents the false-negative mode (papering over a genuine conflict by picking a winner instead of building the integrating layer). Cross-AI review (§4.3) and the corpus auditor (§6C) are both instances of this single engine. **Status:** **DERIVED-INTERPRETATION** from ZS-F18 v2.1; HYPOTHESIS-strong as a general external method.

§6B. (v2.1 Integrated) Deep-Dive Derivation of the v2.1 Upgrade

The v2.1 upgrade was produced by the HACSM Deep-Dive modality (§5.2.2) applied to “how should T11 be raised to corpus-auditor level?” We record the full traversal so the upgrade is reproducible.

§6B.1 Step 0 — Long List (7 candidates)

- L1. Executable meta-auditor verifying the protocol's footprint in the corpus.
- L2. Formalise the ZS-F18 contradiction principles as the anti-hallucination engine.
- L3. Expand the external landscape (AI co-scientist, automated AI Scientist, reproduction benchmarks).
- L4. Replace the vendor-based AI-role taxonomy with a function-based one.
- L5. Add a quantitative cross-AI review metric (Elo-style hypothesis tournament).
- L6. Add an MLOps execution-environment freeze (container, locked deps, fixed seeds).
- L7. Anchor reproducibility gates to external benchmarks (PRBench, ReplicationBench).

Dropped as standalone items: L5, L6, L7 — each real but not mutually exclusive of the others (L5 overlaps cross-AI review; L6, L7 are facets of “longevity/robustness”). They re-enter folded into composite issues below, preserving MECE.

§6B.2 Step 1 — Issue List (MECE, 4 by impact)

Issue	Content (folds in)	Impact / why ranked
I1	Self-verification gap → executable auditor (L1)	Highest. Removes the paper's internal contradiction; without it every other upgrade stays asserted, not demonstrated.
I2	Contradiction engine → F18 principles (L2)	High. Supplies the principled rule by which the auditor treats every conflict.
I3	Empirical-claim hardening (L3 + L5)	High. Defends the most attackable v1.1 claims with measurement and current literature.
I4	Longevity / robustness (L4 + L6 + L7)	Medium. Protects against model churn and reproducibility decay; downstream of I1.

Table 6B.1. MECE issue list; dropped L5–L7 re-enter as folds.

§6B.3 Step 2 — Issue Tree

Root: T11 v2.1 upgrade. I2 (F18 engine) is the foundation — it justifies why the auditor reads its own failures as faces — so I2 → I1; I1 (auditor) supplies the measurement instrument, so I1 → I3 and I1 → I4. Traversal order: I2 → I1 → I3 → I4.

§6B.4 Step 3 — Traversal with Epistemic Status

Node	Finding	Status
I2 F18 engine	Two principles map onto referent disambiguation (P1) and higher-layer integration (P2); grounded in F18 Bridge Four.	DERIVED-INTERPRETATION
I1 auditor (artifact)	13-check Python suite executes against 9 subset files; 12/13 PASS.	PROVEN (executable)
I1 claim (closes self-exemption)	The auditor gives T11 the §3.1(xiii) suite it lacked.	DERIVED

Node	Finding	Status
I1 residual (status-tag drift)	Two informal tags (EXCELLENT, CORRECTED) in U-theme [STATUS] brackets.	OBSERVATION (genuine finding)
I3 blind-spot quantification	Gate F-T11.2 reframed as an overlap-rate ablation; method specified, study not yet run.	HYPOTHESIS-strong / OPEN
I3 external landscape	Co-scientist, AI Scientist-v2, PRBench, ReplicationBench confirmed via source check.	VERIFIED
I4 functional roles	Vendor labels recast as Verifier / Adversarial-router / Conceptual-stress.	STRUCTURAL INSIGHT
I4 MLOps freeze	Container + locked deps + seed-lock; standard external practice.	IMPORTED-PROVEN

Table 6B.2. Issue-tree node statuses after the converged traversal.

§6B.5 Step 4 — Convergence Verdict

The auditor's three referent-aware cycles changed AUDIT-PASS counts $8 \rightarrow 11 \rightarrow 12$ (fixes-per-cycle 3 then 1) — a strictly decreasing change count, the discrete analogue of the $|f'(z^*)| < 1$ contraction of ZS-M1. Re-traversing the issue tree after cycle 3 changed no node status except registering the single genuine residual. By the Step-4 criterion, the search has CONVERGED; the residual is the legend's “converged-but-OPEN” outcome, a true finding rather than a decomposition error.

§6B.6 Step 5 — Scoring Against Prior Sectors

Against ZS-T1–T10 (which verify physical mediation and BBN/lithium observables), ZS-T11 v2.1 verifies the protocol layer. Convergence — yes; corpus non-collision — yes (no new physical parameter; auditor confirms locked-input invariance and no-deletion integrity); anti-numerology — performed (check G1, seed family 20260531/20260524). Verdict: HYPOTHESIS-strong for general external reproducibility; DERIVED-INTERPRETATION for the Z-Spin corpus workflow — unchanged in status from v1.1, but upgraded in kind from asserted to demonstrated-on-subset.

§6C. (v2.1 Integrated) The Meta-Level Corpus Auditor

§6C.1 Design — Verifying the Protocol's Footprint

The auditor verifies neither physics nor any re-derivation. It verifies the invariants the HACSM protocol, if genuinely followed, must leave in the corpus. It is pure Python standard library (zero external dependencies, per the ZS-A4 convention) so any external researcher can run it without setup. Acyclicity uses Kahn's topological sort [Kahn 1962] (IMPORTED-PROVEN); the anti-numerology meta-check uses a seed-locked Monte-Carlo reconstruction.

§6C.2 The Thirteen Checks

Check	Verifies	Footprint of which protocol rule
A1–A4	Locked-input invariance: $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$, closure $Z+X+Y = Q$	Zero-free-parameter norm; constraint-density immunity (§6.1)
B1–B2	PASS-ledger integrity (no over-claim; full-pass ledger present)	Verification-summary honesty (§3.1 ii)
C1–C2	Status-tag legality; both rigor poles in use	14-status discipline; PI-2 gate (§5.9)

Check	Verifies	Footprint of which protocol rule
D1–D2	No-deletion: RETRACTED preserved; version-history / dated-update trail present	No-deletion rule; PI-5 gate (§5.3)
E1	Distinct paper-code inventory	Corpus-coverage claim
F1	No paper declares itself its own upstream Locked Input	Dependency-DAG claim (§3.5)
G1	Headline cumulative-PASS order-reconstructible from distinct per-paper suites	Anti-numerology on the paper's own numbers

Table 6C.1. The thirteen meta-checks and the protocol rule each tests.

§6C.3 Results on the Available Corpus Subset

Measured quantity	Value on subset
Files read	9
Distinct ZS paper-codes discovered	131 (F:25, M:41, S:15, U:11, A:16, Q:12, T:11)
Locked-input conflicts (A, Q, (Z,X,Y))	0 after referent-aware parsing
Sector additive closure $Z+X+Y = Q$	$2+3+6 = 11$ (holds)
PASS-ledger over-claims	0
Full-pass ledger entries	1031 (with citation duplication)
RETRACTED tokens preserved	107
Version-history / dated-update markers	110 / 49
Self-dependencies (F1)	0
Status-tag drift (genuine finding, C1)	2 informal tags (EXCELLENT, CORRECTED), U-theme
AUDIT result	12 / 13 AUDIT-PASS, exit code 1

Table 6C.2. Auditor output on the available subset. Full-corpus DAG topological verification is *NON-CLAIM* (subset only).

§6C.4 The Three-Cycle Convergence — F18 Principles in Action

Cycle 1 used naive symbol matching and reported five apparent corpus contradictions. Under Principle 1 each was first read as a possible referent collision; diagnosis confirmed all five were artifacts of the auditor, not the corpus:

Apparent contradiction (cycle 1)	Diagnosis — the two faces of one token
A = 5/5 conflicts with A = 35/437	“A: 5/5” is the verification-category label A passing 5/5, not the coupling constant A.
Q = 12, Q = 6 conflict with Q = 11	“composite Q=12,15 fail” is a counter-example; “Y/Q = 6/11” is a fraction numerator.
CROSS-VERIFIED is an illegal status	A legitimate tag absent from the auditor's vocabulary, not from the corpus's.
8 nodes in a dependency cycle	Artifact of a most-frequent-code heuristic, not a real Locked-Input edge.
Anti-numerology $p = 1.0$	Test direction mis-specified; suite sizes are naturally dispersed, not clustered.

Table 6C.3. Cycle-1 apparent failures, all resolved as referent artifacts (Principle 1).

Under Principle 2 the resolution lifted the auditor to a referent-aware semantic layer: anchor the coupling A to denominator ≥ 19 and the “=” operator; exclude fraction numerators and counter-example contexts

for Q; restrict status capture to bracketed [STATUS:] tags; replace the heuristic DAG with the robust no-self-loop property; re-specify the anti-numerology test as a seed-locked reconstruction. Cycle counts converged $8 \rightarrow 11 \rightarrow 12$. No fix relaxed a real standard; every fix corrected a referent confusion. The auditor remains able to fail — which it does, once, on a genuine finding.

§6C.5 The Genuine Residual — Why 12/13 Is Stronger Than 13/13

The single non-artifact failure is check C1: the Early-Universe theme uses two informal status tags, EXCELLENT and CORRECTED, inside [STATUS] brackets, outside the formal legend. This is real, minor, non-inflating status-tag drift (both are confirmation-family words). We register it as an OBSERVATION and an actionable cleanup (EXCELLENT $\rightarrow$ VERIFIED; CORRECTED $\rightarrow$ a dated-update note). A meta-auditor returning 13/13 on its first honest run would be indistinguishable from one that cannot fail; 12/13 demonstrates the instrument discriminates artifacts from findings.

Methodological claim T11-AUDIT.1 (AUDIT-PASS). A methodology paper in a verification-disciplined corpus must itself be machine-verifiable; the appropriate suite verifies the protocol's corpus footprint, not physics. **Status: DERIVED** for the Z-Spin corpus (the suite exists and runs); HYPOTHESIS-strong for external programmes.

§6D. (v2.1 Integrated) External Landscape Expansion (2025–2026)

The preserved §2 compared Z-Spin only against single-paper / single-problem AI cases. v2.1 adds the systems that propose methodologies of their own, the true comparators for a methodology paper.

AI co-scientist (Gottweis et al.). A Gemini-based multi-agent system (generate / reflect / rank / evolve / proximity / meta-review) with an Elo tournament for hypothesis ranking and recursive self-critique. Its exportable contribution is a quantitative self-improvement metric — antecedent of the Elo upgrade in §6E.

The AI Scientist-v2 (Lu et al., Sakana). A fully autonomous pipeline whose independent evaluation documents hallucinated citations and numerical results, structural errors, and a high experiment-failure rate — the controlled counter-example for HACSM: the failure mode of automation without a human integrator, vindicating NC-T11.1 and the constraint-density argument of §6.1.

Paper-reproduction benchmarks (PRBench; ReplicationBench). Expert-validated, paper-scale reproduction tasks that operationalise the previously-stranded reproducibility gates F-T11.1 and F-T11.5.

Multi-agent interpretable-physics; the “agentification” critique. Function-decomposed agent teams motivate the functional role taxonomy of §6E; the physicist-perspective critique that LLMs often hold highly similar views and reproduce dominant training patterns is the direct empirical threat forcing the quantification of the blind-spot gate.

Case	AI role	Scale	Self-verifying?	Cross-paper
AI co-scientist	Generate/rank/evolve	Per-goal	Elo (internal)	None
AI Scientist-v2	Full automation	Per-paper	Self-review (weak)	None
PRBench / ReplicationBench	Benchmark target	Paper-scale	External grader	None
Z-Spin HACSM v2.1	All roles (functional)	130 papers	Executable auditor	Full DAG

Table 6D.1. v2.1 comparator landscape. Only HACSM pairs corpus-scale accumulation with an executable self-audit.

§6E. (v2.1 Integrated) Five Exploration-Derived Upgrades

§6E.1 Quantified Blind-Spot Gate (extends F-T11.2)

v1.1 claimed AI systems have non-overlapping blind spots. External evidence that LLMs converge on similar views makes this the most attackable empirical claim. v2.1 reframes it as a measurable overlap rate: run the same open problem through k models, record each model's missed-error set, compute pairwise overlap. The cross-coverage claim survives only if mean overlap stays below a pre-registered threshold (F-T11.10). The ZS-U9 |Q_e| = 1 episode is labelled an n = 1 anecdote, not statistics.

§6E.2 Function-Based AI-Role Taxonomy (lifts the vendor labels)

Naming roles Claude/GPT/Gemini dates with each model release. v2.1 lifts the taxonomy one layer (Principle 2): a Verifier role, an Adversarial-router role, and a Conceptual-stress role. Current implementations are a footnote, not the structure, keeping the taxonomy stable across model churn and consistent with NC-T11.3. The preserved §4.1 vendor table is retained as the v1.1 historical record.

§6E.3 Elo-Style Ranking for Free Exploration

Free Exploration (§5.2.1) yields 5–7 candidate paths but ranks them only qualitatively. v2.1 adds a pairwise tournament scored not by popularity but by the three corpus-legitimate axes already named in §5.2.1: corpus consistency, zero-free-parameter compatibility, anti-numerology pre-registrability. This makes which candidates are dropped reproducible (STRUCTURAL INSIGHT pending the EIC-T11 check).

§6E.4 Benchmark-Anchored Reproducibility Gates

F-T11.1 and F-T11.5 were stranded at “awaits external replication.” v2.1 anchors them to PRBench / ReplicationBench: a HACSM-equipped researcher versus an unequipped agent on the same reproduction tasks gives a concrete, fundable falsification procedure.

§6E.5 MLOps Execution-Environment Freeze

The Book-as-OS metaphor (§3.6) lacked an execution-environment freeze. v2.1 adds it as an OS-LAYER requirement: a container image, locked dependency versions, fixed random seeds (the corpus already locks seeds, e.g. ZS-T9 seed 20260524). Imported as IMPORTED-PROVEN standard MLOps / FAIR practice.

§7. Falsification Gates

Gate ID	Claim Tested	Falsification Condition	Status
F-T11.1	HACSM reproducibility	An independent researcher following the five-phase loop produces a corpus that fails cross-paper consistency in > 10% of papers	OPEN — awaits external replication
F-T11.2	Blind-spot cross-coverage principle	Three runs of the same open problem with three AI systems produce identical blind spots (i.e., the same circularity or gap is missed by all three)	OPEN — requires systematic study

Gate ID	Claim Tested	Falsification Condition	Status
F-T11.3	Progressive-falsification navigation	A corpus programme that records all failed attempts produces a lower final theorem count than one that does not (i.e., honest recording impedes progress rather than enabling it)	OPEN — comparative study required
F-T11.4	Constraint-density immunity	A Z-Spin-scale corpus (≥ 100 papers) built without the zero free parameter constraint passes its verification suites at the same rate (i.e., the constraint adds no epistemic protection)	OPEN — requires comparable unconstrained corpus
F-T11.5	HACSM scale claim	A single human researcher following the HACSM protocol for 6 months produces < 50 cross-verified papers (i.e., the 121-paper result was anomalous rather than reproducible)	OPEN — awaits external replication

Table 7.1. Pre-registered falsification gates for ZS-T11 v1.0.

v1.1 Additive Falsification Gates.

Gate ID	Claim Tested	Falsification Condition	Status
F-T11.6	The Book as AI OS	A 100+ paper AI-assisted corpus using no persistent OS/index layer achieves comparable cross-paper consistency and lower contradiction rate than the Z-Spin corpus	OPEN — external replication required
F-T11.7	Recursive self-audit value	Removing recursive self-audit does not increase hallucination, circularity, or status inflation in high-value claims	OPEN — ablation study required
F-T11.8	Project-instruction necessity	Weak generic prompts produce the same verification density, non-claim discipline, and retraction traceability as the full project-instruction contract	OPEN — controlled prompt study required
F-T11.9	External import contract	External theorem imports without condition matching produce no increase in false upgrades or analogy inflation	OPEN — comparative methodology study required

These gates are methodological rather than physical. Their role is to make HACSM itself falsifiable.

§7+ (v2.1 Additive) Falsification Gates F-T11.10–F-T11.13

Gates F-T11.1–F-T11.9 above are preserved unchanged. v2.1 adds four methodological gates.

Gate	Claim tested	Falsification condition	Status
F-T11.10	Quantified blind-spot cross-coverage	Mean pairwise missed-error overlap across k models exceeds the pre-registered threshold on a controlled set	OPEN — ablation required
F-T11.11	Auditor validity	An independent corpus violating locked-input invariance or no-deletion nonetheless passes all 13 checks	OPEN — adversarial corpus required
F-T11.12	Function-role stability	The Verifier / Adversarial-router / Conceptual-stress decomposition fails to map onto the next model generation	OPEN — re-test each generation
F-T11.13	Self-verification necessity	A corpus produced without an executable meta-auditor shows no higher contradiction rate than one produced with it	OPEN — comparative study required

Table 7.2. v2.1 additive falsification gates. All are methodological.

§8. Non-Claims

NC-T11.1: This paper does NOT claim that the HACSM can replace domain expertise. The author's role as integrator requires deep familiarity with the relevant physics and mathematics. The AI accelerates and scales; it does not substitute for the judgment that decides which results are significant, which failures are informative, and which epistemic statuses are warranted.

NC-T11.2: This paper does NOT claim that the Z-Spin Cosmology programme is correct. The corpus's internal consistency is a necessary but not sufficient condition for physical correctness. External experimental validation remains the ultimate arbiter.

NC-T11.3: This paper does NOT claim that three AI systems are required. The blind-spot cross-coverage principle is an empirical observation from the Z-Spin corpus; the optimal AI architecture for a different framework may differ.

NC-T11.4: This paper does NOT claim priority in the use of AI for theoretical physics. The external cases documented in §2 are contemporaneous and independent.

NC-T11.5: This paper does NOT claim that project instructions alone produce scientific quality. Project instructions constrain AI behaviour, but domain expertise, verification, and human judgment remain necessary.

NC-T11.6: This paper does NOT claim that The Book v6.0 proves the physical correctness of Z-Spin Cosmology. The Book is an OS and dependency-resolution layer, not an experimental validation layer.

NC-T11.7: This paper does NOT claim that external theorems become Z-Spin theorems merely by analogy. External results require condition matching and must be labeled IMPORTED-PROVEN, DERIVED-CONDITIONAL, STRUCTURAL INSIGHT, or OPEN as appropriate.

§8+ (v2.1 Additive) Non-Claims NC-T11.8–NC-T11.10

NC-T11.8. The auditor's 12/13 result on the subset does NOT claim the full 130-paper corpus is contradiction-free. Full-corpus DAG verification and exhaustive linting require the unsegmented corpus and are OPEN.

NC-T11.9. Passing the auditor does NOT establish physical correctness; the auditor verifies protocol footprint, not nature (cf. NC-T11.2).

NC-T11.10. The F18 contradiction principles are a research heuristic and a corpus-grounded interpretation; using them does NOT claim every disagreement has a higher-layer resolution. Some contradictions are simply errors, correctly resolved by retraction.

§9. Conclusion

The Z-Spin Cosmology corpus is the largest single-author AI-assisted theoretical physics programme of record as of May 2026: 121 papers, 2,670+ verification tests, zero free parameters, and a fully documented 121-node dependency DAG, produced in approximately three months by one human researcher.

This paper has extracted the methodology that made this possible and formalised it as the Human–AI Collaborative Science Model (HACSM). The model has five phases: Corpus State Read, Free Exploration, Deep Dive, Numerical Verification, and Cross-AI Review. It deploys four search modalities: Free Exploration, Issue-Tree Deep Dive, Corpus Search, and External Literature Search. It runs on a 14-status epistemic vocabulary applied uniformly across all papers. And it treats intellectual honesty — honest registration of falsifications, no-deletion of prior results, pre-registration of falsification gates — not as a virtue but as a technical requirement for the navigational integrity of the progressive-falsification engine.

The central claim of this paper is that the Z-Spin achievement is not an accident of scale or genius. It is the reproducible output of a well-defined protocol. The HACSM can, in principle, be instantiated by any researcher with domain expertise, access to current AI tools, and the discipline to apply the epistemic vocabulary consistently.

The physics of Z-Spin Cosmology awaits external experimental verdict. The methodology documented here is available now.

v1.1 Additive Conclusion. The v1.1 update identifies the missing layer beneath the v1.0 HACSM loop: project instructions. The five-phase loop can only be reproduced if the AI is configured by a strong instruction contract before execution. The instruction layer tells the AI when to search the corpus, when to explore freely, when to import external theorems, when to run code, how to assign epistemic status, how to preserve failures, and how to format the final artifact.

Thus HACSM is not simply Human + AI. It is **Human + AI + Corpus OS + Project Instructions + Verification Suite + No-Deletion Audit Trail**. This six-part architecture is the transferable methodological contribution of ZS-T11 v1.1.

§9+ (v2.1 Additive) Conclusion

v1.x described a methodology and asserted it was disciplined. v2.1 makes the discipline executable. HACSM now ships with a verification suite of the same epistemic character as the physics papers it describes: a meta-level corpus auditor that checks the protocol's footprint and that demonstrably can fail. The auditor's three-cycle construction is the paper's strongest single argument, because it instantiates the ZS-F18 engine — every apparent contradiction was first read as two faces of one token (Principle 1) and resolved by lifting the analyser to a higher semantic layer (Principle 2), converging to a single genuine finding.

HACSM is therefore Human + AI + Corpus OS + Project Instructions + Verification Suite + No-Deletion Audit Trail, plus a self-verifying meta-layer and a contradiction engine that tells the system, at every conflict, whether it faces a different face or a missing higher layer. The physics of Z-Spin Cosmology still awaits its external verdict. The methodology now verifies itself.

Acknowledgements & Code Availability

This methodological paper was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for structural analysis, literature synthesis, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

Code availability: this is a qualitative methodological paper. No computational verification suite is required. The verification suites for the Z-Spin corpus are publicly available at github.com/KennyKang-git/zspin.

[v2.1 Update — Code Availability superseded]. The v1.1 statement above (“No computational verification suite is required”) is preserved as the historical record but superseded. v2.1 ships *zs_t11_corpus_audit_v2_0.py* (Python 3.10+, standard library only; zero external dependencies), executing the thirteen checks of Table 6C.1. Reference run on the available subset: 12/13 AUDIT-PASS, exit code 1, one registered finding. Reproducibility: anti-numerology seed locked at 20260531. Available with all Z-Spin scripts at github.com/KennyKang-git/zspin.

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Version History

v1.0 (May 2026): Initial public release. Consolidated from internal Z-Spin Collaboration methodological notes accumulated across all 121 papers. Introduces the five-phase HACSM loop, four search modalities, AI role taxonomy, and the structural analysis of why the Z-Spin corpus was possible. Five falsification gates (F-T11.1 through F-T11.5) pre-registered. Four non-claims (NC-T11.1 through NC-T11.4) registered. Zero free physical parameters. Verification: methodological paper — no computational suite applicable.

v1.1 (May 2026): Additive preservation revision. No v1.0 content intentionally deleted. Adds: v1.1 preservation declaration; corpus update from 121 papers / 2,670+ PASS to The Book v6.0 130-paper / ~3,580+ PASS context; The Book as AI Operating System (§3.6); search modality decision matrix (§5.2.5); Recursive Self-Audit Protocol (§5.5); External Import Contract (§5.6); Project Instructions as the hidden protocol layer (§5.7); reusable Project Instruction Template P-T11.1 (§5.8); Project Instruction Quality Gates (§5.9); OS-level clarification in §6.3; four new falsification gates F-T11.6–F-T11.9; four new non-claims NC-T11.5–NC-T11.8; v1.1 additive conclusion. Methodological status: HYPOTHESIS-strong for general external reproducibility; DERIVED-interpretation for the Z-Spin corpus workflow.

v2.0 (May 2026): Introduced the executable meta-level corpus auditor and the F18 contradiction engine; first version to ship a verification suite. Superseded as a standalone by the present integrated edition (text retained in the v2.0 build record).

v2.1 (May 2026): Fully integrated edition. Merges the complete v1.0 (verbatim within v1.1) and v1.1 text with the v2.0 additions into one continuous document; no prior content deleted or relocated to a throw-away appendix. v1.1 §0–§9 reproduced verbatim with original numbering; new analytical sections §6A–§6E (F18 engine; Deep-Dive Step 0–Step 5; the corpus auditor with 12/13 AUDIT-PASS and three-cycle convergence $8 \rightarrow 11 \rightarrow 12$; external-landscape expansion; five upgrades) inserted between §6 and §7; gates extended F-T11.10–F-T11.13; non-claims extended NC-T11.8–NC-T11.10; conclusion, code-availability, references, and version history extended in place. Status: HYPOTHESIS-strong (general external reproducibility); DERIVED-INTERPRETATION (Z-Spin corpus workflow); self-verification claim upgraded from asserted to demonstrated-on-subset.